

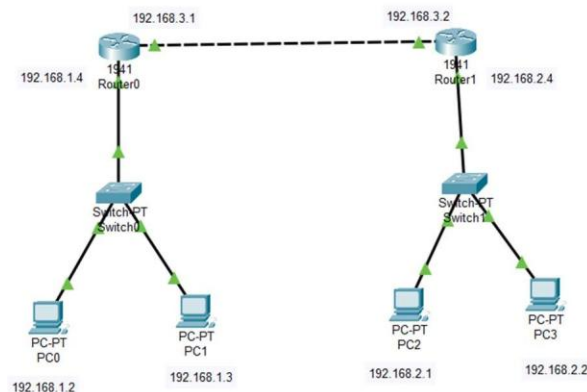
EXPERIMENT NO:-05

AIM:- To study about the configuration of static routing Inter-connection and communication of different networks by using Cisco Packet Tracer.

● STEPS :

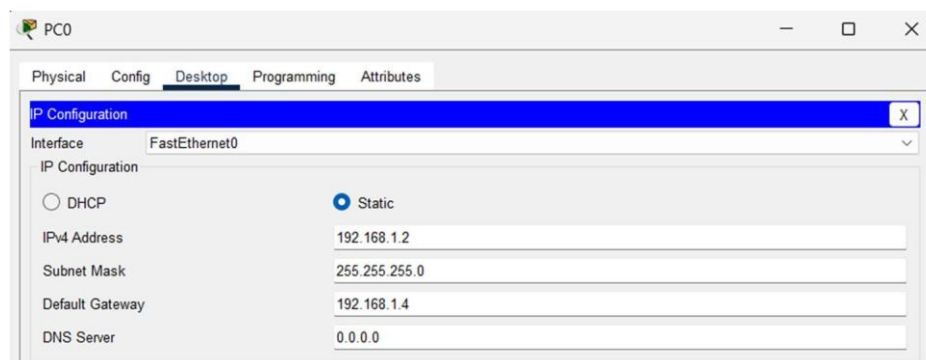
1. Setup the Network:

- Create a connection similar to the one shown in the reference diagram.
- Arrange routers, switches, and PCs, and connect them using appropriate cables.



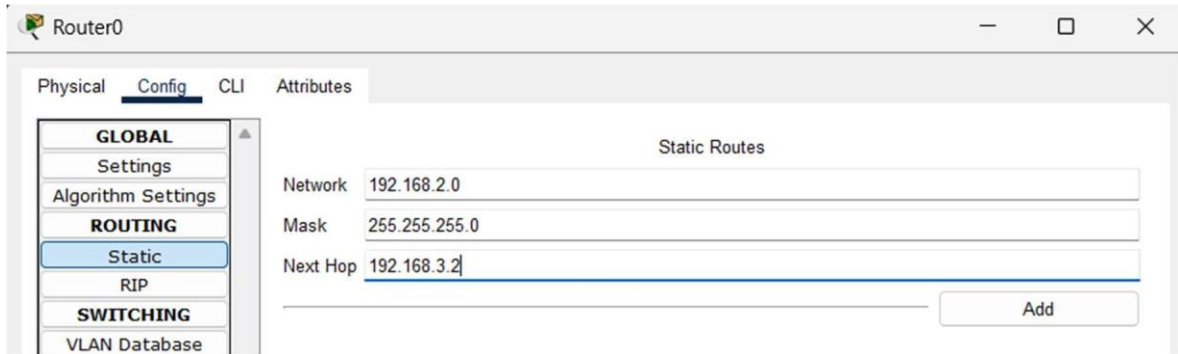
2. Assign IP Addresses:

- Assign IP addresses to all routers and PCs step by step.
- For each PC, go to the **Desktop Tab > IP Configuration** and configure the IPv4 Address, Subnet Mask, and Default Gateway.



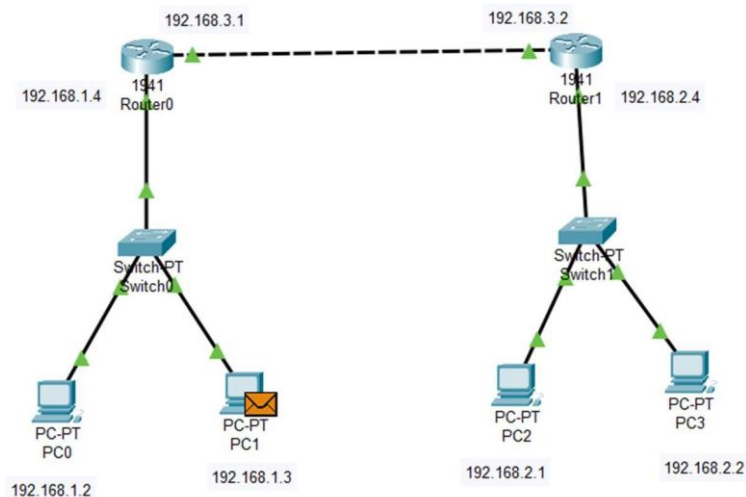
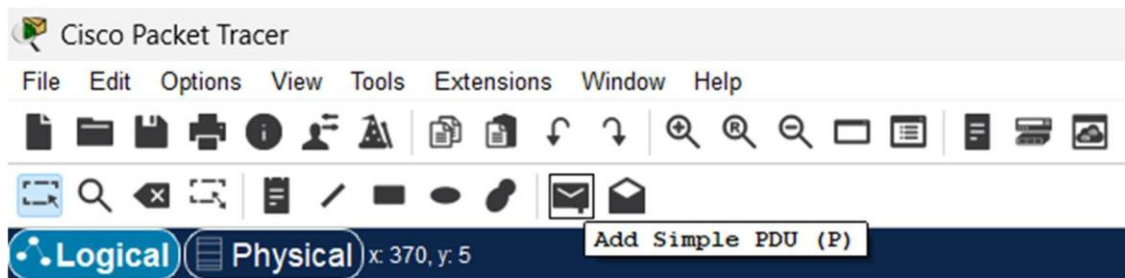
3. Static Routing Configuration:

- Configure static routes on the routers.
- Use the **Network IP Address**, **Subnet Mask (255.255.255.0)**, and **Next Hop Address** (IP of the target router where data needs to be routed).



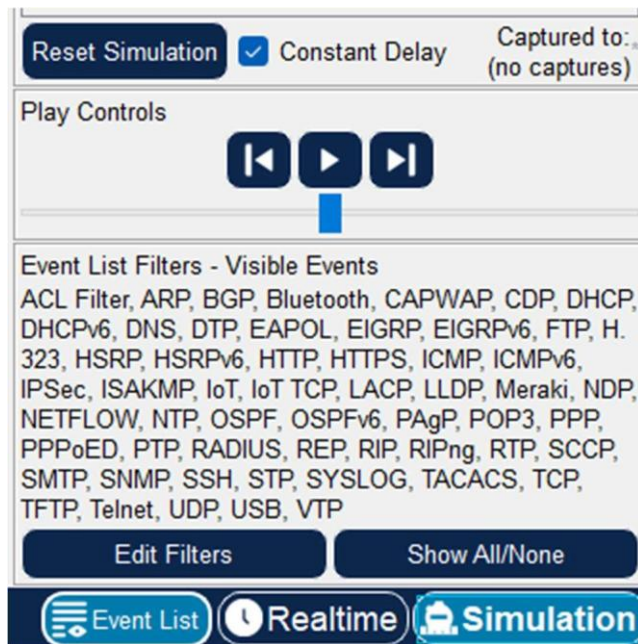
4. Send Data:

- Use the **Add Simple PDU** tool.
- Select a PC and send a message to a PC in a network connected to a different router (not directly connected).



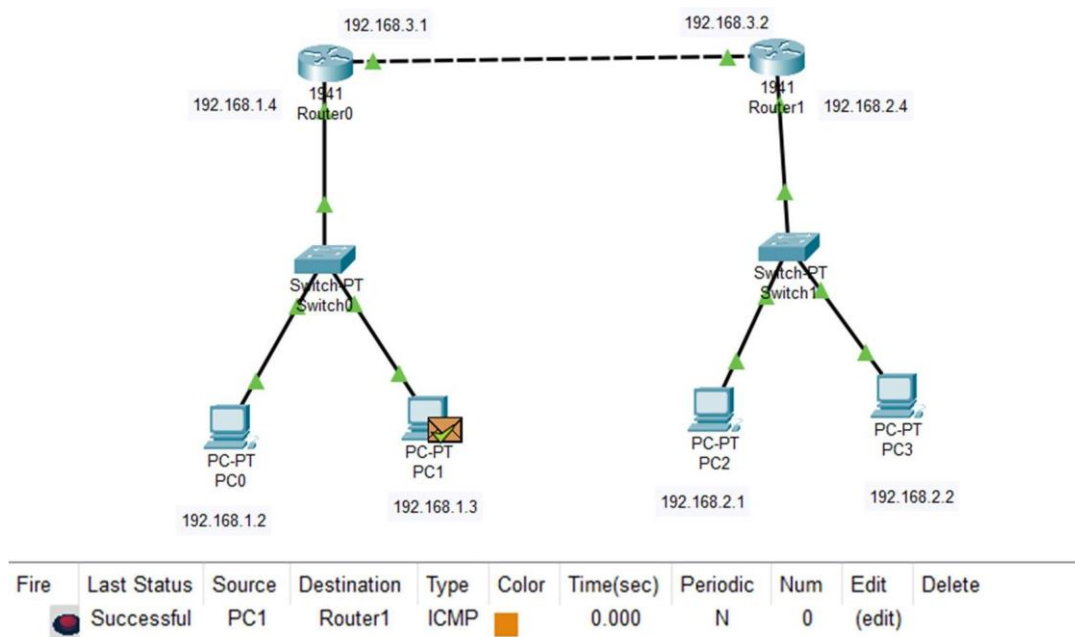
5. Simulation Mode:

- Click on the **Simulation Mode** located at the bottom right.
- Play the simulation to visualize the routing process and check the communication flow.
- Event List.



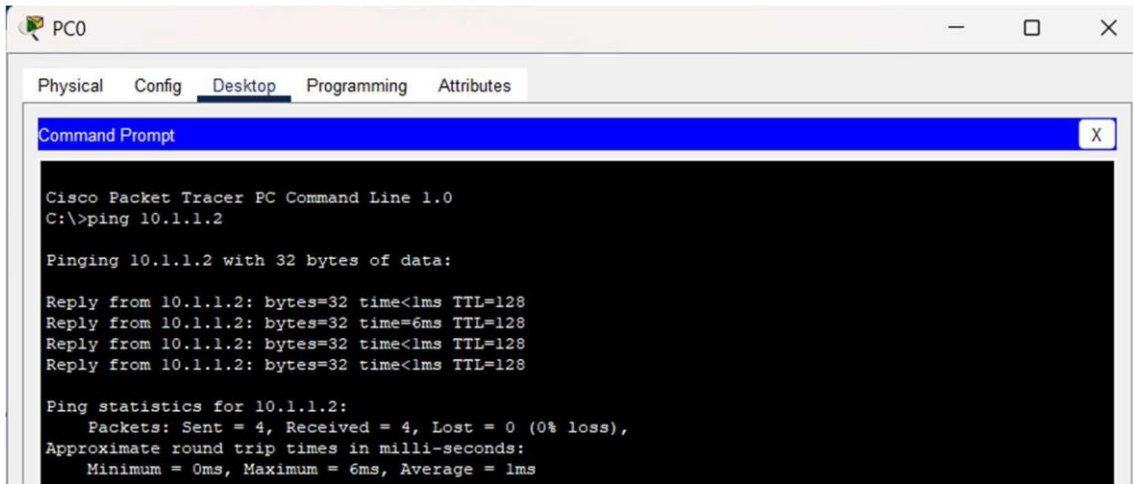
Vis.	Time(sec)	Last Device
	0.000	--
	0.001	PC1
	0.002	Switch0
	0.003	Router0
	0.004	Router1
	0.005	Router0
	0.006	Switch0

6. After it done successfully we see like in shown the picture final step.



7. Verification:

- Open the **Command Prompt** from the Desktop tab of any PC.
- Use commands like `ping` to verify the communication between devices on different networks.



The screenshot shows a Cisco Packet Tracer PC Command Line window for a PC named PC0. The window has tabs for Physical, Config, Desktop, Programming, and Attributes, with Desktop selected. The Command Prompt window displays the following text:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.1.1.2

Pinging 10.1.1.2 with 32 bytes of data:

Reply from 10.1.1.2: bytes=32 time<1ms TTL=128
Reply from 10.1.1.2: bytes=32 time=6ms TTL=128
Reply from 10.1.1.2: bytes=32 time<1ms TTL=128
Reply from 10.1.1.2: bytes=32 time<1ms TTL=128

Ping statistics for 10.1.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 6ms, Average = 1ms
```